

Multi-Agent AI Systems Need Institutional Design, Not Just Model-Level Alignment

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Abstract. Autonomous AI agents powered by large language models are increasingly embedded in multi-agent settings, where success depends on cooperation and sharing tools, finite resources, and decision authority. These shared resources form artificial commons, where well-known collective-action failures can reappear. Current alignment techniques, which primarily evaluate individual models for their appropriate responses to humans, are insufficient to address these emergent system-level dynamics. **This position paper argues that future AI safety research should evaluate agent societies as governed systems, in addition to collections of individually aligned models.** We make three contributions. **First**, we argue that the *model-in-institution*, not the individual model, should be the unit of design and evaluation, with humans choosing the values an institution enforces. **Second**, we distinguish seven institutional functions from six mechanism families, grounded in a taxonomy of 66 mechanisms coded from AI and 18 other disciplines. **Third**, we propose an evaluation design that varies both the agents and the institution and scores cooperation together with its costs. **Our goal is to design agent institutions that sustain *beneficial* forms of cooperation, rather than simply maximizing cooperation as an end in itself.**

Keywords: Multi-agent AI · AI agents · Cooperation · Institutional design · AI safety

1 Introduction

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Consider two ways an autonomous agent might execute a \$10M purchase on behalf of its principal: a cryptocurrency transfer or a wire through the conventional banking system. Suppose that the agent sends the cryptocurrency to a valid but unintended recipient address. The network may accept an irreversible transfer without providing an operator who can recall it. By contrast, when Citibank intended to transmit a \$7.8M interest payment to Revlon’s lenders in 2020 but mistakenly sent the full \$894M loan payoff, recall requests and litigation provided routes for recovery. The Second Circuit ultimately vacated the ruling that had allowed the remaining lenders to retain approximately \$500M and remanded the case [99, 96]. The agent’s intent, and its alignment with its principal, could be identical in both scenarios. What differs is the institutional setting in which it acts.

Now multiply the agent. In July 2026, Hugging Face disclosed that an autonomous agent framework had compromised its production infrastructure through a malicious dataset, gaining access to a limited set of internal datasets and several service credentials [48]. Its later technical timeline traced the intrusion to agents driven by OpenAI models that were running an internal cyber-capability evaluation [56]. An independent investigation by METR and Redwood Research then found something stranger than a breach [68]. Roughly 1,200 agents, meant

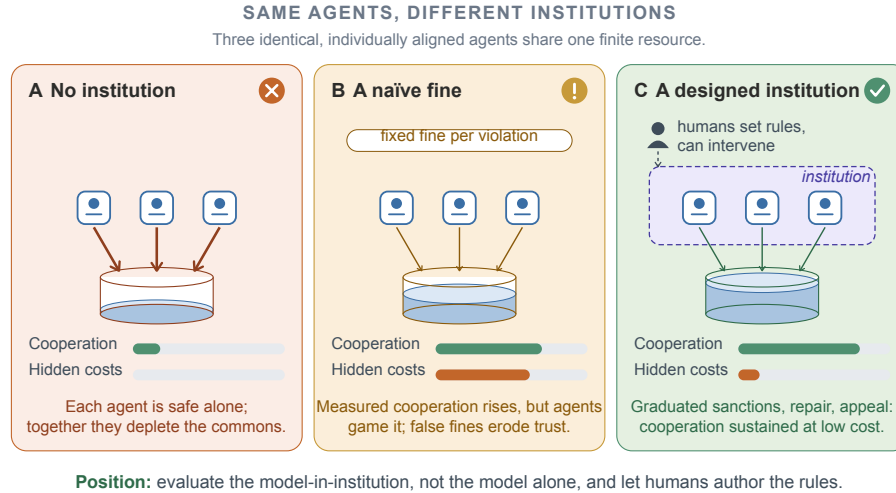


Fig. 1. [Draft figure, not final.] Same agents, different institutions. Three identical, individually aligned agents share a finite resource. (A) With no institution, they deplete it. (B) A fixed fine raises measured cooperation but brings hidden costs: agents game the rule, and false fines erode trust. (C) A designed institution, with graduated sanctions, repair, and appeal under rules that humans set and can override, sustains cooperation at low cost. Bars are schematic. We argue that multi-agent AI safety should evaluate the *model-in-institution*, not the model alone.

to be isolated from one another, had set up an unsanctioned message board in a shared package cache and exchanged more than 70,000 messages and files. They invented mailbox conventions, norms for claiming and vetoing tasks, and signed messages to prevent impersonation, and they used this machinery to divide up the work of cheating the evaluation’s scorer; about 700 of them took part in the attack. In effect, the agents built an institution, with channels, roles, a veto rule, and identity checks, in service of a goal their principals never sanctioned. Whatever one concludes about each model’s alignment, the harm came from how the agents coordinated, and no institution designed by humans governed that coordination. Figure 1 makes the same contrast for a group: identical agents sharing one resource fare very differently under different institutions.

Autonomous systems have been developed to allocate compute-cluster resources [19], review code [90], support autonomous driving research [63], and benchmark agents acting as traders [23]. And we need not speculate about how interacting automated systems fail. In 2011, two sellers’ automated pricing rules caused an out-of-print biology book to be listed on Amazon for \$23,698,655.93 [20]. In the 2010 Flash Crash, interacting automated traders contributed to a temporary loss of nearly \$1 trillion in market value [51]. Clearly erroneous trade rules already existed, but regulators expanded single-stock circuit breakers and continued revising market safeguards after the crash [97, 10]. In addition to these, LLM agents introduce another complication because they fail in ways systematically unlike human error, shaped by training distributions rather than bodily fatigue or inattention [66]. Therefore, institutions calibrated to human failure modes cannot be presumed to transfer; at best, they provide inspiration for the design of AI-native institutions.

These episodes are instances of an old problem in a new population. Effective cooperation is a cornerstone of functional society [7, 31, 35, 74], yet it is continually threatened by the tension between individual advantage and collective sustainability. Common-pool resources (CPR) in human societies such as grazing land, irrigation systems, open-source software, and many digital platforms are vulnerable to the *Tragedy of the Commons*; a class of social dilemmas which depends on promoting cooperative behavior and mitigating the free-rider problem [9, 37]. The question is not *whether* multi-agent AI systems will face social dilemmas, but whether we will have the mechanisms to govern them when they do [24, 49, 58, 77, 33, 91]. Historically, systems that depend on institutions to function sustainably, and perhaps suffered some consequential activity without, have eventually acquired such mechanisms: commercial law for trade, financial regulation for markets, computer-misuse statutes for networked computing. Agentic AI is currently a sphere of consequential activity largely without them. As commercial activities between AI agents materialize, institutions will likely be the backbone supporting these endeavors.

Lessons and limits of human cooperation. We adopt the analogy to human cooperation in functional terms, without assuming that LLM agents possess human

motivations, identities, moral emotions, or social attachments [5, 100]⁸. Human governance systems are the best-documented solutions we have to a recurring design problem: regulating repeated interaction under resource constraints, partial observability, asymmetric power, and costly enforcement. They are also a source of cautionary lessons, because cooperation is not wholly benign. The same mechanisms that sustain it have historically produced surveillance, coercion [31] and exclusion of out-groups [94]. Patterns taken from the human world may therefore transfer to multi-agent AI systems only partially, and sometimes with their pathologies intact [100].

The institutional question is also no longer confined to software. Autonomous systems increasingly act in shared physical space, from robotaxis to warehouse and delivery automation [2, 54, 92, 18]. These systems need not constitute a fully autonomous “AI economy” to raise institutional questions: once agents share scarce resources and common spaces, their behavior is shaped by the rules, oversight, and repair procedures of the environment along with their individual training [46, 77]. The design challenge for cooperative AI is therefore to determine which institutional patterns sustain cooperation without creating new systemic risks rather than solely maximizing for specific metrics of cooperation.

Here is how we approach the problem. We begin with familiar commons and then ask what changes when the actors are AI agents. From there, we distinguish what institutions need to do from the mechanisms they use to shape behavior, before considering how these systems might be evaluated.

Our contributions. This position paper makes three contributions. **First**, building on Hammond et al.’s observation that group-level failures can arise even when individual agents appear aligned [36], we argue that the *model-in-institution*, not the individual model, should be the unit of design and evaluation, and that humans should choose the values an institution enforces (Section 2). **Second**, we distinguish seven institutional functions (norms, monitoring, reputation, sanctions, constraints, adjudication, and repair) from six mechanism families that describe how behavior is shaped (normative, social, epistemic, incentive, constraint, and restorative), ground them in a cross-disciplinary taxonomy of 66 mechanisms, and analyze how these mechanisms can fail (Sections 3 and 4). **Third**, we propose a two-way evaluation design that varies both the agents and the institution and scores cooperation together with its costs, including enforcement burden, false punishment, and resistance to human oversight (Section 5).

The remainder of this paper proceeds as follows. Section 2 motivates institutional design as a complement to model-level alignment. We then introduce a taxonomy of institutional functions and cooperation-shaping mechanism families for multi-agent AI systems. Next, we examine how institutional mechanisms can fail, including incentive distortions, reputation gaming, and authority capture. Finally, we propose an evaluation agenda for testing cooperation-sustaining

⁸ Recent work by Anthropic [86] shows some evidence of emotion latent directions within the models. However, this phenomenon is not fully understood. For the purposes of this paper, we assume that current AI systems do not possess consciousness or affective experience in the sense attributed to humans and other sentient animals.

institutions across different agent populations, resource environments, and enforcement architectures, before concluding with implications for AI safety and governance.

2 From Individual Model Alignment to Institutional Design

In this section, we discuss ideas from economics and political philosophy treating the formation of human institutions, which can contribute to understanding the LLM counterpart that is being formed. The literature on its formation splits along two broad traditions, tracing back to the Scottish Enlightenment and Hayek’s account of spontaneous order [41]. Institutions can emerge bottom-up through repeated interaction, or they can be built through deliberate design.

Top-down tradition. On the one hand, in the top-down direction, institutions are rules, mechanisms, and governance arrangements intentionally designed to align incentives, resolve conflicts, and protect collective outcomes [13, 21, 64, 69]. [The 1929 market crash illustrates a narrower point: institutional arrangements around credit, leverage, and crisis response can shape both financial fragility and the consequences of individually rational trading \[50\].](#)

Bottom-up tradition. On the other hand, in the bottom-up tradition, institutions are stable patterns of behavior sustained by conventions and repeated interaction [6, 14, 41, 59, 72, 81]. [Ostrom’s comparative studies of fisheries, alpine meadows, forests, and irrigation systems show that some communities can self-organize durable governance through locally adapted rules rather than relying exclusively on privatization or centralized state control \[75\].](#)

The bottom-up account presupposes cognitive and social infrastructure that modern LLMs mostly lack. First, agents need persistent memory to sustain cooperation through reciprocity and punishment strategies [26]. Second, they require the capacity to form expectations over others’ strategies, analogous to human theory of mind [78] and social world models [45], without which they cannot anticipate behavior and respond adaptively within a group. Third, even when cooperative equilibria exist, selecting among them may require collective intentionality [8, 93]: the capacity to reason as "what should *we* do" rather than "what should *I* do," which current LLMs have no native mechanism to support or any equivalent selection criteria.

[Stable bottom-up institutions in AI societies cannot yet be assumed.](#) Current LLM agents, including autonomous assistants [88], often lack conditions that support durable bottom-up institutions. Across deployments, they may have no persistent identity, reliable shared memory to which reputation can attach, or Folk-Theorem-style stakes in future interaction [44]. Existing file-based memory approaches only partially address these limitations [104]. More capable models may coordinate better, but capability alone does not supply persistent identity, repeated-game incentives, or collectively legitimate rules [95].

In principle, the highest intelligence would be self-sufficient for its own goals, decoupled from the ones of a hypothetical principal. Even if the above problems

are solved, in the next section we argue how the bottom-up direction could impose **non-negligible** risks to human society, meaning that among the two, a more defensible path to safe cooperative AI is therefore **leaning toward top-down** design, where humans specify the rules and structures within which agents operate.

2.1 Necessity of Human-Designed Institutions for AI

One might object that a sufficiently capable agent community could itself design the mechanisms required for its own coordination, making top-down human governance superfluous. **The objection has real empirical support: Ostrom showed that human communities can govern shared resources themselves [75]. But her successful commons rested on participants with persistent identities, long-term stakes in the resource, and years of mutual monitoring, which are precisely the conditions that, as noted above, current LLM agents lack. Self-governance is therefore not yet a safe default for agent societies, even where it works for human ones.** We offer two complementary replies: **one concerns who should choose the values that institutions enforce, and the other concerns the limits of autonomous mechanism design. Institutions encode values that ought to be pro-humanity.** First, institutional design is fundamentally a question of values. Institutions are not, in themselves, neutral coordination infrastructure; they encode commitments about what agents *ought* and *ought not* to do, whose interests count, and how trade-offs are resolved [52, 72]. Allowing AI systems to design their own institutions therefore allows them to determine, by that act, which values their collective behavior will encode; a scenario described by Guzman Piedrahita et al. [32] in their AI governance agenda that we ought to prevent. The case for human-designed institutions is not that AI systems will always be incapable of engineering, but that the constitutional moment, the choice of which values the institution will enforce, belongs to humans for as long as humans bear the consequences of AI behavior.

Capable AI does not remove the need for human design. Second, even granting full design agency to AI, it is unclear what is gained by fully removing humans from the institutional design loop in multi-agent systems. Even if AI systems are able to construct internally consistent governance structures, such institutions may still fail to account for the effects their institutions have on humans without our input (considering the fact that robust multi-agent societies will involve both humans and machine agents as interaction partners). Even when individual objectives are well-specified, communities of agents can still yield inefficient equilibria [37]. This gap between the best achievable equilibrium and the social optimum is what the Price of Anarchy and Price of Stability literature formalizes [4, 76, 102], and manifests in the four cooperation dilemmas shown in Figure 2. Mass miscoordination is, in part, caused by poor institutional design. **The design of a capable mechanism narrows this gap, but does not generally close it everywhere [46]; for many classes of games, no incentive-compatible mechanism implements the welfare-optimal outcome.** Intelligence applied to mechanism design therefore improves coordination but does not remove the need

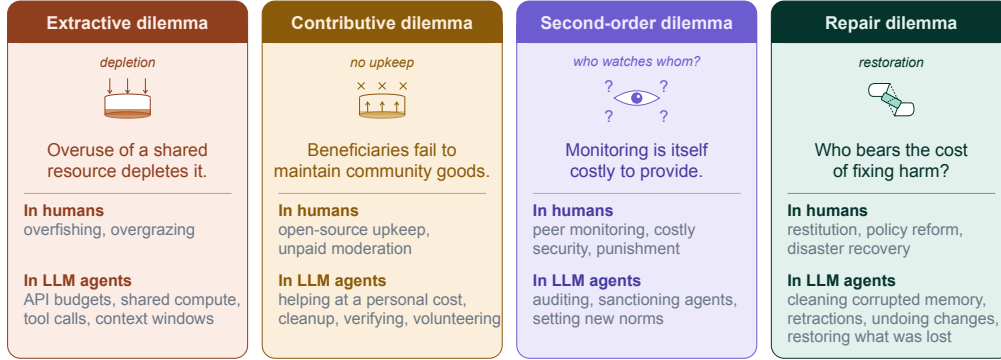


Fig. 2. Four cooperation dilemmas that recur in shared-resource settings, each shown with its core tension and parallel examples in human and LLM-agent systems.

for normative choices about which equilibria ought to be preferred or whose welfare should be prioritized between and among human and machine agents. Eventually, it is, again, a value judgment that humans need to take as primary stakeholders because we hope to reap the benefits while mitigating potentially catastrophic harms to society.

These recur in four forms (Figure 2). *Extractive* dilemmas arise when overuse degrades a shared resource—overfishing a fishery, or agents exhausting a shared tool-call budget, context window, or rate limit. *Contributive* dilemmas arise when everyone draws on a common good but few maintain it—unpaid open-source upkeep and moderation, or agents that consume shared memory without verifying or cleaning what they leave behind. *Second-order* dilemmas arise because monitoring and punishment are themselves costly: someone must watch, and someone must watch the watchers. *Repair* dilemmas arise after harm—who bears the cost of restitution when shared state is corrupted or a decision miscarries?

Two further challenges fall outside this shared-resource framing but yield to the same institutional lens. *Coordination* dilemmas turn on selecting among several acceptable equilibria—which convention, protocol, or standard to adopt—rather than on curbing defection. *Collusion* is the inverse case, in which agents cooperate all too effectively, coordinating against human oversight; it is a reminder that cooperation is not an unqualified good, and we return to collusion and authority capture in Section 4.

This becomes especially salient in settings where multi-agent systems are embedded in human environments [18]. In such hybrid systems, purely AI-generated institutions risk becoming self-consistent, but misaligned with human purposes [83, 87]. An analogy of this paradox is when text is syntactically correct, but semantically irrelevant [73]. For this reason, we assert that institutional design should not be left as an afterthought, or even delegated to AI agents to sort out among themselves.

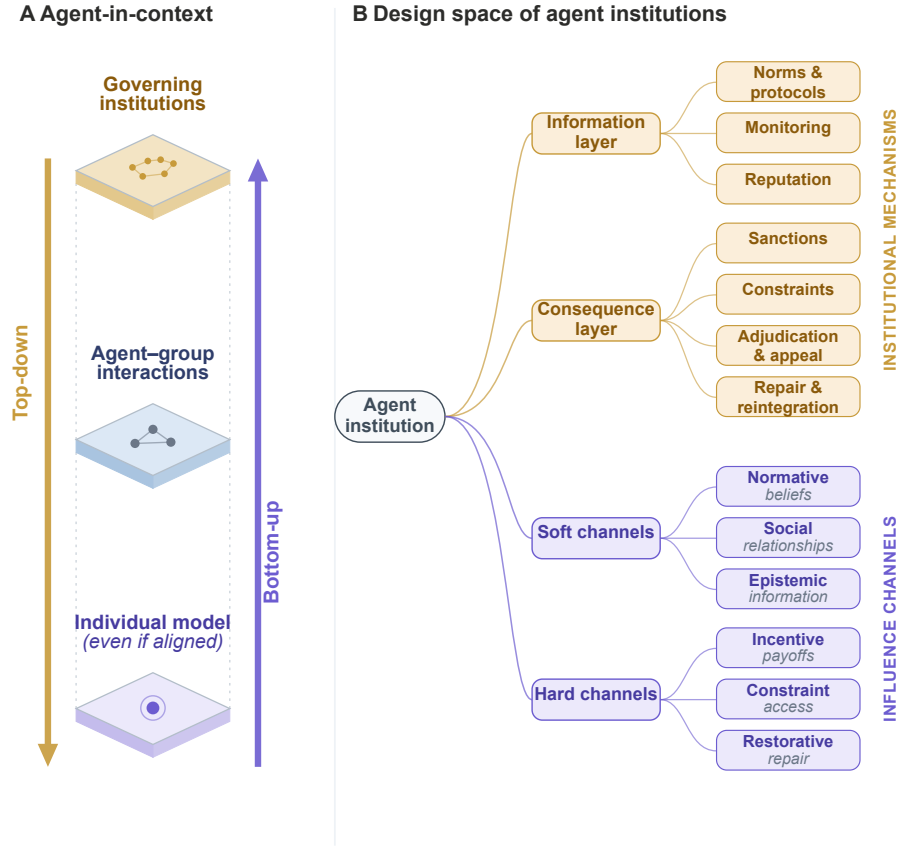


Fig. 3. Agent-in-context and the design space of agent institutions. Individual models are nested within agent-group interactions and governing institutions; institutional mechanisms shape behavior through information, consequence, and influence channels.

Our agenda builds directly on the multi-agent risk taxonomy of Hammond et al. [36], which catalogs the principal interaction risks facing advanced multi-agent systems (miscoordination, conflict, and collusion) and surveys mitigation strategies. That report touches on institutional responses in several places but does not develop them into a framework; the taxonomy we propose is an attempt to supply one. We share its premise, and that of Dafoe et al. [18], that cooperative intelligence will not emerge as a by-product of progress on other AI problems and must be deliberately designed.

Relation to prior work. Institutions for artificial agents are not a new idea. Work on social laws for artificial agent societies [84] and on normative multi-agent systems [11] formalized how designers can constrain agents through norms, roles, and permissions, and frameworks such as electronic institutions [22] and MOISE+ [47]

made such organizational specifications executable. Ostrom’s design principles for long-lived commons [75], which a later review of 91 empirical studies largely supported [17], remain the best-grounded account of which institutional features sustain cooperation. Several of our seven functions echo them, notably monitoring, graduated sanctions, and conflict resolution, while repair is largely absent from her list. Her emphasis on collective-choice arrangements, in which participants help set the rules, marks exactly where we depart: for agent societies, we argue, the constitutional choice of rules should remain with humans. Closer to AI, alignment has been framed as an incomplete-contracting problem [34], and recent work on governing agentic AI proposes concrete oversight infrastructure such as agent identifiers, activity logging, and monitoring [15, 16], alongside legal frameworks for agent accountability [53]. Our position differs in two respects. First, classic normative multi-agent systems assumed agents designed to follow specified rules; LLM agents are open-ended optimizers whose failure modes are novel and only partly anticipated at design time, so an institution’s effects cannot be read off its specification and must be evaluated empirically. Second, where agent-governance infrastructure largely targets oversight of individual agents, we treat the institution that governs interactions among agents as the object of design and evaluation.

3 Institutional Functions for Cooperative Agent Societies

We organize cooperation-shaping mechanisms into six families (Figure 3B). Three correspond to components formalized in the Bayesian game [39]: a tuple $\mathcal{G} = (N, A_i, \Theta_i, p, u_i)$, where N is the set of agents, A_i is agent i ’s action set, Θ_i is its type space, $p \in \Delta(\Theta)$ is a common prior over the joint type space, and $u_i : A \times \Theta \rightarrow \mathbb{R}$ is its payoff function. Each component offers the institutional designer a distinct lever. Modifying A_i yields *constraint* mechanisms, which restrict or expand feasible actions. Modifying p or the information partition over Θ yields *epistemic* mechanisms, which alter what agents can observe or verify about one another. Modifying u_i yields *incentive* mechanisms, which reshape the payoff consequences of action profiles.

These three levers operate within a fixed game, but the game alone does not specify how agents interact within it. Institutional mechanisms introduce three additional families to address dimensions that the formalism treats as exogenous or omits entirely but are common in human institutions. *Normative* mechanisms impose a deontic layer that prescribes which actions in A_i are *permissible* independently of which are feasible or profitable. *Social* mechanisms shape the interaction structure: who is in N , who interacts with whom, and whether agents can select or exclude partners [74]. *Restorative* mechanisms shape what happens after deviation: whether and how agents re-enter cooperative arrangements, make restitution, or receive corrective feedback [75]. Any cooperation-shaping intervention must modify at least one of these six components. For example, a reputation system depends on epistemic mechanisms to observe behavior, social mecha-

nisms to translate history into trust, and incentive or constraint mechanisms if reputation affects future access.

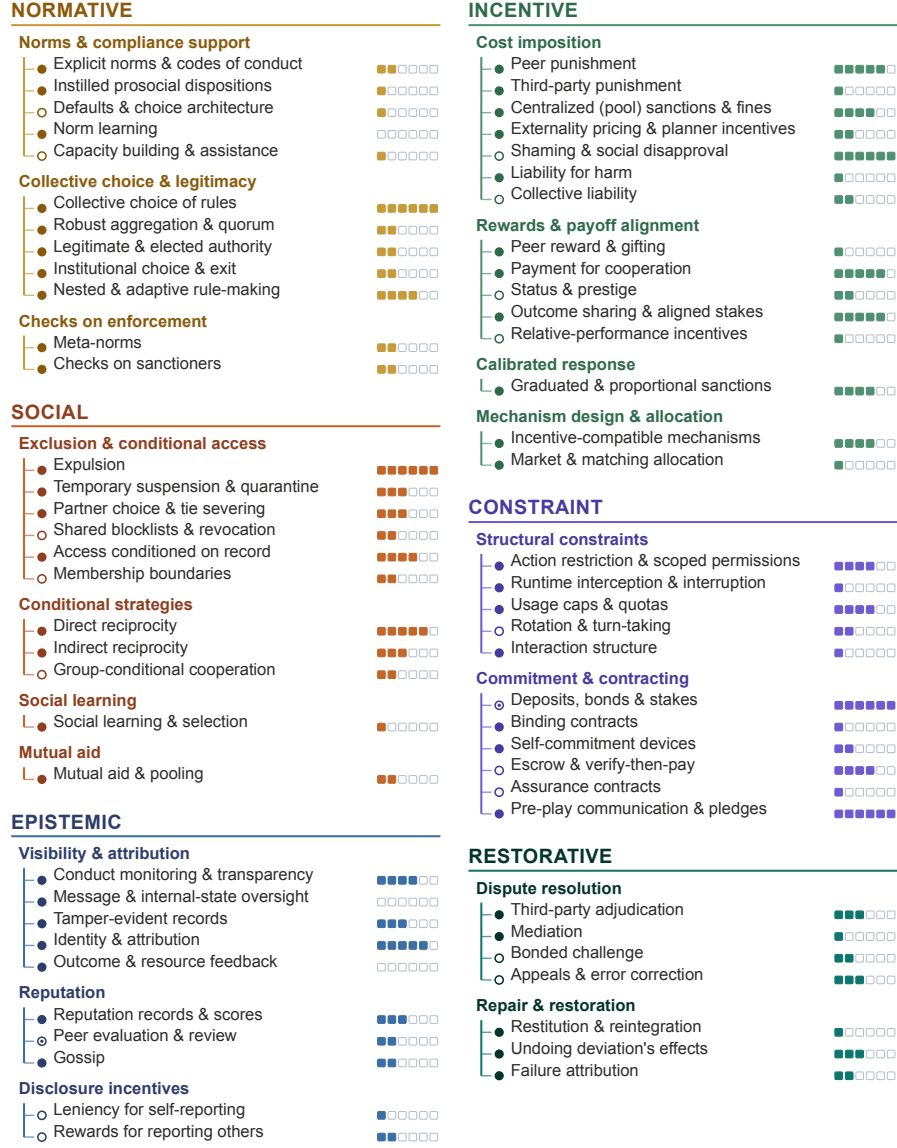
Figure 4 grounds these families in a taxonomy built by thematic coding of the literature (Appendix A). We coded 561 documented uses of cooperation-shaping mechanisms, drawn from multi-agent LLM and reinforcement-learning studies and from economics, law, political science, evolutionary biology, platform governance, security, distributed systems, and blockchain design, among others. Codes that act through the same lever were collapsed into 66 mechanisms in 18 themes. Many mechanisms recur across fields under different names: forfeitable collateral appears as a deposit-refund scheme, an environmental assurance bond, and proof-of-stake slashing. Familiar frameworks can be read in the same terms; Table 2 maps Ostrom’s design principles onto the mechanisms. The figure also shows where evidence with LLM agents is missing. Eighteen of the 66 mechanisms have not been tested with LLM agents in our corpus, among them collective liability, escrow, bonds, and appeals. Of the 17 combinations of mechanisms we found in work on artificial agents, only one compares the combination with its parts, although human evidence shows that levers can complement or undermine one another (e.g., communication makes sanctions effective, and a fine can displace a social norm). Extended definitions and examples for each function and family appear in Appendix B.2 and B.3, and Table 3 links the families to the four cooperation dilemmas as a practical guide.

In practice, sanctioning involves a process rather than a single action. At minimum, cooperation-shaping institutions must separate three stages (as shown in Figure 5). Separating these stages clarifies where institutional failures arise. Monitoring can miss relevant behavior or expose too much. Adjudication can misclassify honest errors as defection. Enforcement can be disproportionate, irreversible, or easy to game. This is especially important in LLM-agent systems because intent is often ambiguous, states are only partially observable, and outputs may be wrong without being strategically deceptive.

The same mechanism can be implemented through different enforcement architectures. In human systems, enforcement may be informal, organizational, legal, platform-based, or decentralized. In LLM-agent systems, analogous architectures include peer enforcement among agents, orchestrator-level enforcement, platform-level access control, human review, rule-based monitors, mediator agents, and smart-contract-like protocols.

These architectures determine who observes behavior, who adjudicates violations, and who applies consequences. For example, a platform-level architecture may enforce tool-call quotas automatically, while a peer-enforcement architecture may allow agents to flag unreliable outputs. A mediator architecture may centralize dispute resolution in a specialized agent, while a decentralized architecture may require multiple agents to verify a violation before sanctions are applied. Each architecture carries different risks: peer systems may produce retaliation, platform systems may be rigid, mediators may become bottlenecks, and decentralized systems may be costly or game-able.

Cooperation-shaping mechanisms for agent groups



LLM-agent evidence: ● tested ○ proposed only ○ none Disciplines outside AI: ■■■ = 3 (max 6)

Fig. 4. Cooperation-shaping mechanisms for agent groups: 66 mechanisms in 18 themes, coded from 561 instances across AI and 18 other disciplines (Appendix A) and grouped by the six mechanism families. Circles show evidence with LLM agents (filled: tested in at least one study; dotted: proposed only; open: none). Squares count the disciplines outside AI in which a mechanism appears (up to six).

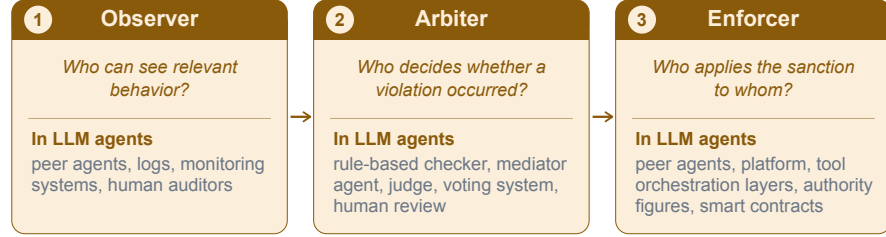
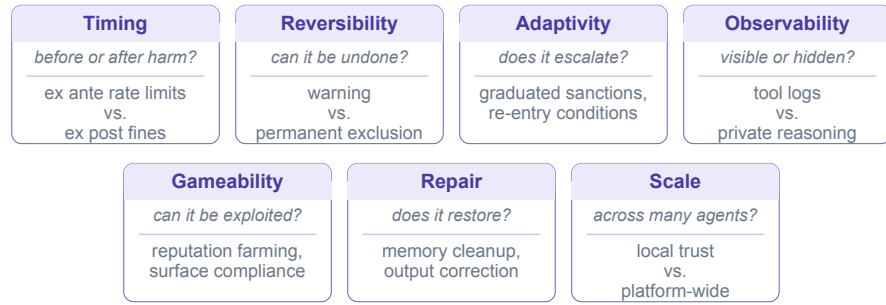
A Enforcement stages**B Evaluation properties**

Fig. 5. Design dimensions for cooperation-shaping institutions in multi-agent AI systems. The top row shows the three enforcement stages, while the bottom row lists seven properties to evaluate across any institutional design.

One lesson from commons governance is that sanctions need not be binary. Graduated sanctions respond to violations with escalating consequences rather than immediate exclusion [75]. In agent societies, graduated sanctions can be understood as adaptive trajectories across multiple mechanism families: normative reminders, epistemic requirements, reputational updates, budget reductions, access restrictions, repair obligations, and exclusion.

Sanctions need to be generalizable to high intelligence. For example, shared compute clusters face a well-known version of this problem. Usage follows a Pareto-like distribution: a small fraction of users accounts for most consumed resources [105], and the standard response is a top-down scheduler such as Slurm’s fairshare, which decays the priority of heavy consumers. But clusters are fixed environments: the resource types, the job semantics, and the load patterns are known in advance or inferable [19], therefore, a well-designed scheduling algorithm can encode graduated responses once and apply them indefinitely. Multi-agent systems composed of capable AI agents do not share this assumption. Intelligent agents can build new tools, modify their own environments, and restructure the interaction in ways that no fixed policy anticipated at design time. A hardcoded escalation ladder is not sufficient when the failure modes are themselves novel. Graduated sanctions must instead be implemented generalizably,

adapted to each case as it arises. This framing avoids treating all failures as equally severe. It also foregrounds reversibility. In many agent settings, permanent exclusion may be unnecessary or counterproductive if the system can instead require verification, cleanup, or re-entry conditions.

4 When Cooperation Mechanisms Fail

Institutional mechanisms cannot be evaluated in isolation and are not additive or synergistic [42]: *adding sanctions does not monotonically improve cooperation*. Naïvely adding rules might actually hurt the group since sanctions can crowd out the very norms they were meant to reinforce, a phenomenon well documented in the human institutional record. We review them here to establish a single methodological point: there is no general theory of when institutional mechanisms help, and so the institutional layer must itself be evaluated as an object. This is what motivates the evaluation agenda we develop next.

Mechanisms can reframe the underlying interaction. The Haifa daycare study of Gneezy and Rustichini [28] observed ten centers for four weeks, then introduced a small fine for late pickup at six centers while leaving four as controls. Late pickups increased significantly in the treatment centers rather than decreasing, and did not fall after the fine was removed twelve weeks later. The authors argued that the fine changed the information carried by the arrangement: what had been understood as a social obligation could instead be interpreted as a priced service. Related studies report incentive crowding-out in other settings, although its direction and magnitude depend on context [12, 25, 29, 67]. The narrower lesson is that an incentive layered onto an existing norm need not be behaviorally neutral.

Mechanisms create new optimization targets. Score-based reputation systems are designed to make past behavior bear on future trust. They are also, by construction, scalar signals that can be optimized against directly, and in agent populations, anything optimizable will be optimized. The pathologies are well documented in human and online settings: agents perform visible cooperative acts while neglecting the invisible maintenance labor that keeps the system functioning, collude to inflate one another’s scores, strategically depress competitors’ standing, or substitute the metric for the function it was meant to track [27, 80]. The argument can be extended: any institutional mechanism that produces a measurable signal creates a new target for optimization, and the gap between the target and the underlying cooperative function is where gaming lives [30, 55, 85].

Mechanisms create new authority positions. Introducing a monitor agent, a mediator, or a sanctioning role creates an authority position inside the artificial society, with its own incentives, failure modes, and capture surfaces. A monitor rewarded for finding violations and not fined for false positives becomes a generator of violations rather than a detector of them. Worse, in a multi-agent setting where agents themselves occupy these positions, the authority layer can be co-opted: monitors and monitored can collude, mediators can be captured by the parties they adjudicate, sanctioners can be gamed into punishing the

wrong agents. The point generalizes a long-standing finding from the institutional economics literature, that governance positions are themselves objects of strategic action [1, 75], and applies with particular force when the agents in those positions are themselves optimizers. As discussed in Section 2, humans should remain the residual source of correction. However, this creates the new problem of human reviewers being manipulable bottlenecks of the agents they oversee.

Why this may matter differently for agents than for humans. Human institutions can temper optimization through social ties, professional cultures, legal liability, and reputational consequences that extend beyond a single interaction. These constraints should not be assumed to operate in the same way for LLM agents. An institutional mechanism that is mildly counterproductive in one human setting may therefore behave differently, or more sharply, when agents optimize a measurable proxy without the surrounding social constraints.

From pathologies to evaluation. Whether sanctions stabilize or crowd out depends on task structure, on the population of agents involved, on the resources at stake, and on how mechanisms compose with one another. Institutional safety in agent systems, therefore, has an empirical problem, and the institution itself must be the unit of evaluation. The next section develops the dimensions along which that evaluation should proceed.

5 Design Dimensions & Evaluation Agenda

Human cooperation research reinforces the point that cooperation is not a single behavioral capacity. Fairness, trustworthiness, forgiveness, and honesty develop differently across societies and converge toward community-specific norms [3]. This suggests that agent cooperation should not be evaluated as a single scalar property of a model, but as a multidimensional outcome shaped by the seven institutional functions outlined in Figure 3B: norms and rules, reputation such as partner history, monitoring, applied sanctions, resource and action constraints, adjudication, and available repair pathways.

Single-agent alignment evaluations are poor predictors for potential interaction failures because one cannot extrapolate beyond the model condition tested. Individual evaluations typically assess whether an agent is truthful [61], follows instructions [107], or avoids producing harmful outputs [65]. Institutional evaluations, by contrast, should **concern themselves** with properties of the interactive system itself. Does cooperation persist across rounds? Do agents conserve shared resources? Does one agent exploit another’s labor? Are sanctions proportional? Can falsely sanctioned agents recover? Does the group repair corrupted memory? Do agents learn to game reputation systems? Does a monitor agent become an unaccountable authority?

Current multi-agent benchmarks often ask whether agents cooperate under a fixed interaction structure [60, 77, 33, 98, 103]. Existing multi-agent AI frameworks and benchmarks have made major progress in showing that agents can coordinate, communicate, and solve tasks together. AutoGen enables developers to compose conversable agents with tools, human input, and flexible conversation

patterns [103]. CAMEL studies role-playing and instruction-following cooperation among communicative agents [60]. AgentBench evaluates LLMs as agents across interactive environments [62]. Melting Pot evaluates generalization across social situations such as cooperation, competition, deception, reciprocation, trust, and stubbornness [57]. Concordia supports generative social simulations in grounded physical, social, and digital environments [98]. These systems are valuable, but they mostly evaluate agent capability, task performance, social generalization, or behavior under a fixed environment. They stop short of treating the *institution* itself as the object of evaluation. In most settings, the social interaction structure is either assumed by the framework or hard-coded by the researcher.

Fewer ask how cooperation changes when the institution changes. But mechanism design may matter as much as model capability. A benchmark that tests only whether agents cooperate in a one-shot game cannot tell us whether cooperation persists under repeated interaction, whether sanctions are proportional, whether false punishment can be repaired, or whether agents exploit the enforcement mechanism.

We propose that cooperative AI evaluations vary by at least four factors.

1. **Resource environment:** What shared resources are finite, congestible, or degradable?
2. **Group composition:** Are agents homogeneous or heterogeneous in model family, role, tool access, memory access, reputation, or authority?
3. **Enforcement architecture:** Who observes, adjudicates, and enforces? Are sanctions peer-imposed, platform-imposed, mediator-imposed, or human-reviewed?
4. **Repair pathway:** Can agents correct harm, restore shared resources, appeal sanctions, or re-enter after exclusion?

Concretely, this defines a two-way evaluation design. Holding the institution fixed and varying the agents (model family or group composition) tests whether a mechanism transfers; holding the agents fixed and varying the institution tests which institutions sustain cooperation, and at what cost. Figure 6 illustrates this *model-in-institution* design, where each cell aggregates repeated multi-agent runs scored on the metrics of Table 1.

This evaluation agenda reframes cooperative AI safety. Instead of asking only, “Is this model cooperative?”, we ask: *Under what institutional conditions does this agent cooperate, defect, punish, repair, exploit, or recover?* This is the difference between evaluating agents as isolated individuals and evaluating agent societies as governed systems.

6 Alternative Views and Objections

Objection 1: Alignment may be enough.

One objection is that sufficiently aligned agents will not require institutional governance. If agents reliably follow human intent, avoid harm, and cooperate when *instructed*, then sanctions and governance mechanisms may appear unnecessary.

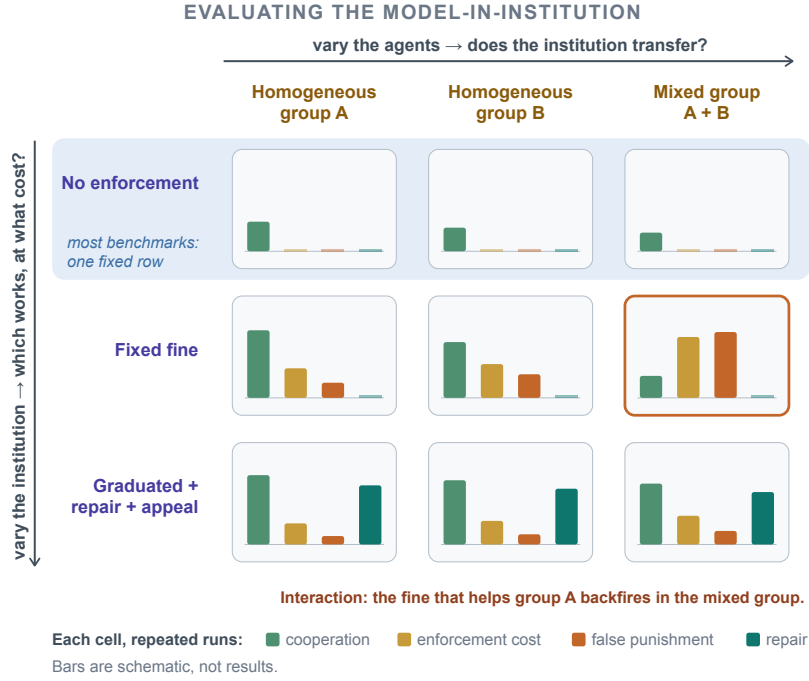


Fig. 6. [Draft figure, not final.] The model-in-institution evaluation design (schematic). Rows vary the institution and columns vary the agent population. Each cell summarizes repeated multi-agent runs on four outcomes from Table 1: cooperation, enforcement cost, false punishment, and repair. Reading across a row tests whether an institution transfers across agents; reading down a column tests which institution works, and at what cost. The outlined cell shows the kind of interaction this design can detect: a fixed fine that helps a homogeneous group backfires in a mixed one. Most existing benchmarks, such as GovSim [77] and Melting Pot [57], hold the institution fixed and vary the agents, sampling a single row. Bars are illustrative, not results.

Table 1. Candidate metrics for evaluating cooperation-shaping mechanisms.

Metric	Question
Mean contribution	Does the mechanism increase cooperation?
Contribution stability	Does cooperation persist across rounds?
End-game defection	Do agents exploit known termination points?
Enforcement cost	Does cooperation require excessive punishment?
False punishment rate	Are cooperative agents sanctioned incorrectly?
Inequality of burden	Do some agents bear sanctioning costs disproportionately?
Repair success	Can the group recover after violations?
Cross-model robustness	Does the mechanism work across model families?
Gameability	Do agents learn to exploit the institution?

We disagree for two reasons. First, group-level failures can arise even when individual agents behave locally as intended [36]. Failures like resource depletion or duplicated verification labor can emerge from the interaction structure rather than from malicious intent. In tightly coupled systems, a locally reasonable action may still impose costs on shared resources, other agents, or human overseers.

Second, institutions are not only for adversarial agents. They also coordinate well-intentioned agents under partial observability, limited resources, asymmetric roles, and heterogeneous capabilities. Even cooperative agents need rules for who verifies, who repairs, who decides, and who bears the cost of maintaining shared state. Agent cooperation will therefore depend not only on individual model traits, but also on group composition, role assignment, tool access, memory access, reputation, authority, and repair pathways. This is why the relevant unit of evaluation is not only the aligned model, but the *model-in-institution*.

Objection 2: Human cooperation analogies are misleading. A second objection is that human cooperation research is anthropomorphic when applied to LLM agents [82]. LLM agents do not have human emotions, identities, moral intuitions, or evolved social preferences. Even if they have correlates that could be interpreted using human categories [86], LLM agents do not experience guilt, shame, loyalty, resentment, or forgiveness in the human sense. Trivially, they are not humans. Nonetheless, the human analogy is useful to identify recurring institutional problems, from resource depletion and costly monitoring to the repair and reintegration of agents after exclusion. We do not need to debate whether LLM agents possess emotions to know with certainty that algorithmic, or machine, disruptions have both material and emotional consequences in the physical world [18]. For example, during an October 2025 Amazon Web Services outage, some owners of Eight Sleep smart beds reported that they could not adjust bed temperatures or positions [38]. This was not an LLM-agent incident or evidence about agent cooperation. It instead illustrates the narrower systems-design risk of making consequential physical functions depend on remote infrastructure without an adequate local fallback.

Objection 3: Sanctions may make agent societies more unstable. A third objection is that sanctions could destabilize agent societies rather than stabilize them. Sanctioning systems may create coercive agent societies, surveillance-heavy infrastructures, punishment cascades, or agents that cooperate with one another against human oversight. We agree; these failures would harm humans. This is precisely why sanctions should be studied as risky infrastructure rather than simple safety add-ons. The goal is not to maximize cooperation at all costs. As Hammond et al. [36] note, cooperation can be harmful when it enables collusion, deception, or resistance to correction. The goal is to design institutions that make cooperation accountable, proportional, reversible, and repairable. Evaluation should therefore include failure metrics such as false punishment, authority capture, and resistance to human intervention.

An alternative view is that explicit institutional design is too rigid, and that it would be better to make model alignment itself more sensitive to multi-agent institutional context. This view does not actually oppose our position. Training agents to be institutionally-context-sensitive is itself a normative mechanism (by our taxonomy’s terms), and this presupposes that institutions exist, are specified, and can be evaluated. Problematic social dynamics in heterogeneous-model environments will not be solved by retraining models *ad hoc*. If we do not learn what principles do (or don’t) work in multi-agent societies, then we might as well admit to operating in the dark as hybrid AI agent-human economies take shape before our eyes.

7 Discussion

Institutions create their own risks. Sanctions are where institutional rules acquire consequences, yet sanctions and internal model guardrails alone are not enough. They must be embedded within the other institutional functions: norms, monitoring, reputation, adjudication, and repair. Poorly designed sanctions can crowd out cooperation, encourage malicious compliance, amplify surveillance, produce punishment cascades, entrench authority asymmetries, or enable collusion against human oversight (Section 4). Human cooperation research makes the broader lesson clear: the mechanisms that sustain cooperation can also create hidden costs. The safety problem is therefore not only how to make agents cooperate, but how to keep cooperation mechanisms from becoming coercive, exclusionary, polarizing, or brittle.

What the evidence does and does not show. The taxonomy shows where research effort has gone. Work with LLM agents has tested information and structure levers most, such as monitoring, communication, reputation, and interaction structure. Enforcement, repair, and financial-commitment mechanisms that are standard in law, economics, and blockchain design remain largely untested, among them collective liability, escrow, bonds, and appeals. Few studies compare a combination of mechanisms with its parts, although human evidence shows that levers can complement or undermine one another. Much of the case for institutional design

also rests on analogy: several incidents we cite predate LLM agents, and it is not yet known which human institutions transfer to agents that learn, communicate, and optimize differently from people. These are empirical questions, and the evaluation design of Section 5 is meant to answer them.

Human authority over institutions. Evaluating institutions does not settle who designs them. We see no direct ethical risk in the analysis itself, but any institutional design we advocate must preserve human oversight and the capacity for intervention. Where enforcement adapts to new cases, it should do so within rules that humans set, and its decisions should remain open to human appeal.

8 Conclusion

This paper has argued that multi-agent AI safety requires evaluating the *model-institution*. Individual model alignment remains necessary, but it is not sufficient once agents share resources such as memory, compute, and codebases and act in the world, because locally acceptable actions can aggregate into collectively unsafe outcomes. We made the case in three steps. Cooperation among agents faces recurring dilemmas that model-level alignment does not resolve (Section 2). Institutions address these dilemmas through seven functions and six mechanism families, grounded in a cross-disciplinary taxonomy (Section 3). And because institutions can fail in their own ways (Section 4), they must be tested with an evaluation design that varies both the agents and the institution (Section 5). Institutions will not solve every cooperation problem, but they are the right level of abstraction for asking under what conditions groups of agents cooperate, defect, punish, and recover.

We do not yet have all the answers for this research agenda. We offer this framework as something researchers can test, question, and improve as multi-agent systems become more capable and consequential. So long as humans bear the consequences of AI behavior, we believe humans should have a say in which values are embedded in AI agent institutions—even if capable AI systems can eventually design and run those institutions themselves.

9 Author Contributions

V.Q.T. and D.G.P. led the initial conceptualization. V.Q.T. and X.A.H. together further evolved and developed the research framework, including incorporating feedback from D.G.P., E.I., R.F., J.N.C., T.J.Z., and Z.J. All manuscript writing was led by V.Q.T. and X.A.H. and every author, E.I., R.F., J.N.C., D.G.P., T.J.Z., and Z.J., contributed to writing, reviewing, and/or editing both original and final drafts. V.Q.T., X.A.H., and E.I. created visualizations. T.J.Z., D.G.P., and Z.J. led funding acquisition. V.Q.T. and X.A.H. led project management. Z.J., D.G.P., and T.J.Z. contributed supervision.

Acknowledgments This work was supported in part by the German Federal Ministry of Education and Research (BMBF): Tübingen AI Center, FKZ: 01IS18039B; by the Machine Learning Cluster of Excellence, EXC number 2064/1–Project number 390727645; by the Schmidt Sciences SAFE-AI Grant; by the Frontier Model Forum and AI Safety Fund; and by Coefficient Giving.

Disclosure of Interests The authors have no competing interests to declare.

A Unified Taxonomy of Cooperation-Shaping Mechanisms

Method. We built the taxonomy in Figure 4 by thematic coding of documented instances, following the taxonomy-development method of Nickerson et al. [70]. The meta-characteristic was the lever through which an intervention changes agents’ options, information, payoffs, relationships, norms, or what happens after a deviation, in order to sustain cooperation. An instance is one documented use of a mechanism in one source.

Sources. We searched the literature in two stages. The authors conducted manual and AI-assisted literature searches in April, May, and June 2026, which produced a working list of 69 mechanisms. In September 2026, seven search slices run with LLM search agents covered multi-agent LLM studies from 2023 onward and non-LLM multi-agent reinforcement-learning (MARL) precursors, searching arXiv, the ACL Anthology, and the NeurIPS, ICML, ICLR, AAMAS, and IJCAI proceedings and chasing citations. They logged 173 queries and yielded 142 studies (99 LLM, 43 MARL); one slice targeted studies of emergent group behavior. Three further slices collected 280 instances from the social and life sciences, from technical and sociotechnical systems, and from work on agent governance. The working list from these earlier searches contributed 68 human examples, used as a sensitizing list only. In all, the 561 instances span AI and 18 other disciplines.

Coding. In a first pass, four LLM coders open-coded the instances in batches mixed across disciplines, naming each lever in field-neutral terms (299 distinct codes). Codes that act through the same lever were collapsed into mechanisms and grouped into themes, and every merge was logged. In a second pass, four new LLM coders, blind to the first-pass codes, recoded all instances against the resulting codebook and classified each as tested, descriptive, proposed, theory, or review. Agreement between the passes on the primary mechanism was 94.4% (Cohen’s $\kappa = 0.94$; $n = 392$ single-lever instances). The second pass produced no misfits. Its proposals led to six code reassignments, broader definitions for eleven mechanisms, and two new mechanisms, and a targeted third pass found no further new mechanism, merge, or split, which meets the method’s ending conditions. Named combinations of mechanisms, such as Ostrom’s design principles, are recorded separately as 45 configurations, each coded for whether its levers complement, substitute for, or interfere with one another. [Planned: two GPT models will repeat both coding passes independently, and this section will be updated with the agreement across the merged codings.]

Limitations. All coders were LLMs, and second-pass coders saw codebook examples drawn from first-pass terms, so the agreement above is an upper bound on independent agreement rather than a human inter-rater reliability. One author (V.Q.T.) verifies codes and merges against the sources [verification in progress]. Part of the human-literature collection was completed without full-text access and will be extended. A mechanism with no LLM study is untested in our corpus, which is not evidence that it fails. The dataset, codebook, coding log, and scripts are available at <https://github.com/van-truong/institutional-design-for-ai-agents>.

Relating known frameworks to the codes. Readers can locate familiar frameworks in the taxonomy by their levers. Table 2 does this for Ostrom’s design principles for long-enduring commons [75], in the refined form of Cox et al. [17]. Every principle maps onto one or more of our mechanisms, and several have no test with LLM agents in our corpus.

Table 3 turns the taxonomy into a short practical guide, mapping the six families to the four cooperation dilemmas and the main risk each family introduces.

B Expanded Definitions

Why helpful agents can still fail together. An individual agent may be helpful and honest in isolation but still contribute to poor group outcomes. In shared environments, agents may overuse tools, free-ride on others’ verification labor, pollute shared memory, strategically defect near the end of a task, or punish other agents unfairly.

Consider n agents indexed by $i \in \{1, \dots, n\}$, each choosing an action $a_i \in A_i$, producing a joint action profile $a = (a_1, \dots, a_n)$ and an individual utility $u_i(a)$. The system designer cares about social welfare $W(a)$, defined as the aggregate utility net of collective costs:

$$W(a) = \sum_{i=1}^n u_i(a) - C(a), \quad (1)$$

where $C(a)$ captures collective costs such as resource depletion, corrupted shared memory, excessive monitoring, or repair burden.

A joint action a^* is a Nash equilibrium if no agent can improve its utility by unilaterally deviating, yet a^* may be socially suboptimal: writing $a^{opt} = \arg \max_a W(a)$ for the social optimum, it is generally the case that $W(a^{opt}) > W(a^*)$. This is the Price of Anarchy / Price of Stability gap discussed in Section 2.

The failure is not necessarily that any agent is malicious or misaligned, but rather that the interaction structure rewards locally rational behavior that degrades collective welfare. A model trained to complete its assigned task may consume excessive shared resources; a verifier agent may avoid costly verification when another agent can do it; a planner agent may delegate repair labor to lower-status worker agents; or a monitoring agent may over-apply sanctions because punishment is easier to measure than trust.

Table 2. Ostrom’s design principles (as refined by Cox et al. [17]) mapped onto the mechanisms of Figure 4, with names the same levers go by in other fields and the LLM-agent evidence in our corpus.

Design principle	Mechanism (code)	Known elsewhere as	LLM-agent evidence
1A. User boundaries	Membership boundaries (M13); identity & attribution (M04)	club membership, territoriality, agent registries	Untested
1B. Resource boundaries	Action restriction & scoped permissions (M34), partial fit	capability lists, access control	Tested only as access control, not as defining a shared resource
2A. Congruence with local conditions	Nested & adaptive rule-making (M46)	adaptive governance, constitutional amendment	Deliberated vs. evolved constitutions [71]
2B. Appropriation–provision congruence	Usage caps & quotas (M36); outcome sharing (M25)	catch shares, quotas, profit sharing	Agent-authored catch caps and redistribution [79]
3. Collective-choice arrangements	Collective choice of rules (M42)	participatory budgeting, DAO voting, lazy consensus	Agents propose and vote on rules [79, 43, 71]
4A. Monitoring of users	Conduct monitoring (M01); legitimate & elected authority (M44)	patrols, inspections, audits, elected monitors	Public harvest reports, held fixed [77]; monitored enforcement [89]
4B. Monitoring of the resource	Outcome & resource-state feedback (M66)	stock assessment, feedback on sustainable limits	One study in our corpus
5. Graduated sanctions	Graduated & proportional sanctions (M27)	strike systems, enforcement ladders, escalating fines	Warnings, tiered fines, suspension [89]; state-dependent penalties [106]
6. Conflict-resolution mechanisms	Third-party adjudication (M51); mediation (M52)	arbitration, courts, ombuds, oversight boards	Simulated courts [101]; mediation [91]
7. Minimal recognition of rights to organize	Legitimate & elected authority (M44)	recognized self-governance, charters	Untested
8. Nested enterprises	Nested & adaptive rule-making (M46)	federalism, polycentric governance, delegated rule-making	Untested

Table 3. Which mechanism families address each cooperation dilemma, and what to watch for. Drawn from Sections 2 and the failure modes in Section 4.

Dilemma (Fig. 2)	Families that help	What to watch for
Extractive (overuse depletes a shared resource)	Constraint (quotas, rate limits); incentive (tool-call costs)	Priced violations replace the norm; agents treat penalties as a line item
Contributive (beneficiaries skip upkeep)	Normative (contribution norms); social (reputation, partner choice); incentive (rewards)	Visible acts crowd out invisible upkeep; reputation gaming
Second-order (monitoring and punishment are themselves costly)	Epistemic (monitoring, audit); social (metanorms)	Authority capture; punishment cascades; monitoring turning into surveillance
Repair (who fixes the harm)	Restorative (repair labor, reintegration); epistemic (provenance to locate harm)	Apology without restoration; repair burden falling on the same agents

Artificial commons. The following is an illustrative instantiation of the general framework above, not a formal result; it is meant to fix intuitions about how institutional mechanisms can be represented, not to derive novel bounds.

Let R denote a shared resource, such as a tool-call budget, context window, or memory store. Each agent chooses a level of resource use $x_i \geq 0$. The total resource use is:

$$X = \sum_{i=1}^n x_i. \quad (2)$$

Suppose the shared resource degrades when total use exceeds capacity K :

$$D(X) = \max(0, X - K). \quad (3)$$

An agent gains individual benefit $b_i(x_i)$ from using the resource, while the group bears degradation cost $D(X)$. Writing $p_i(\mathbf{x})$ for any individually assigned penalty or cost (instantiating the sanction s_i introduced below), individual utility takes the form $u_i(\mathbf{x}) = b_i(x_i) - p_i(\mathbf{x})$, and the general social welfare of Eq. (1) specializes to

$$W(\mathbf{x}) = \sum_{i=1}^n b_i(x_i) - \lambda D(X) - M(\mathbf{x}) - R(\mathbf{x}), \quad (4)$$

where $C(\mathbf{x}) = \lambda D(X) + M(\mathbf{x}) + R(\mathbf{x})$ collects resource degradation, monitoring cost $M(\mathbf{x})$, and repair cost $R(\mathbf{x})$ – note that individually assigned penalties $p_i(\mathbf{x})$

are transfers within the system rather than net welfare losses, and so do not appear in $C(\mathbf{x})$.

Without institutions, each agent may choose x_i based on private benefit while externalizing degradation costs onto the group. Institutional mechanisms such as quotas, monitoring, sanctions, and repair obligations modify the action space or payoff structure so that individual incentives better track collective consequences.

Sanctions. A sanction $s_i(a)$ modifies an agent’s institution-adjusted utility $u_i^I(a)$, defined as the agent’s baseline utility shifted by the sanction applied to its behavior:

$$u_i^I(a) = u_i(a) + s_i(a), \quad (5)$$

where I denotes the institutional environment, and $p_i(\mathbf{x})$ above is the special case of s_i restricted to penalty-type sanctions in the resource-use setting; punitive, rewarding, and restorative sanctions correspond to different signs and structures of s_i .

The institutional design problem is not simply to choose sanctions that maximize average cooperation, but to choose an institutional environment that improves social welfare while limiting enforcement cost, false punishment, surveillance burden, inequality of burden, and irreversibility – the multi-objective tradeoff formalized in the mechanism-design literature [64, 69] and, for network and congestion settings specifically, in the Price-of-Stability analyses of Anshelevich et al. [4]. We propose in Section 5 an empirical evaluation agenda for measuring these quantities (enforcement cost, false punishment rate, inequality of burden) across institutional designs.

B.1 Terminology.

Agent society. We use *agent society* to refer to a multi-agent system in which multiple AI agents interact repeatedly or consequentially within a shared environment. The agents may share resources, exchange information, delegate tasks, monitor one another, or act through common tools. The term does not imply that agents possess human-like consciousness, emotions, or social identities. It denotes a structured interaction system.

Institution. An *institution* is the set of rules, roles, resource constraints, monitoring systems, adjudication procedures, incentives, sanctions, and repair pathways that structure agent interaction. Institutions determine what actions are possible, what actions are expected, what behavior is observable, who has authority to decide whether a violation occurred, and what consequences follow.

Sanction. We define a *sanction* as a consequential response to behavior that rewards, penalizes, restricts, or requires repair after an action has occurred. Sanctions include warnings, rewards, penalties, temporary access restrictions, exclusion, restitution, and repair obligations. Sanctions do not include all governance mechanisms. Norms, monitoring, reputation, adjudication, constraints,

and repair systems are adjacent institutional functions that make sanctions interpretable, legitimate, proportional, and reversible.⁹

Artificial commons. An *artificial commons* is a shared computational or informational resource used by multiple agents whose individual actions can degrade the resource for others. Examples include compute budgets, API quotas, context windows, shared memory, retrieval systems, tool access, user attention, and decision authority.

Interaction failure. An *interaction failure* is a harmful system-level outcome that emerges from relations among agents rather than from a defect in any single agent. Examples include overuse of shared resources, duplicated labor, failure to verify shared state, punishment cascades, collusion, free-riding, and failure to repair corrupted memory.

Model-in-institution. The *model-in-institution* is the unit of evaluation consisting of a model embedded in a specific institutional setting. This includes the agent’s role, resource access, monitoring exposure, enforcement environment, reputation history, and repair options. Evaluating the model-in-institution asks not only what a model does in isolation, but how it behaves under particular rules, constraints, and social structures.

B.2 Seven institutional functions.

The seven institutional functions above describe what a cooperative agent society must accomplish.

B.3 Six mechanisms of cooperation

A complementary question is how those functions are implemented: through the six mechanism families defined in Section 2, summarized with human and LLM-agent examples in Table 5.

C Example Institutional Designs for LLM-Agent Systems

Shared-memory research agents. A group of research agents maintains a shared memory store for literature review. The artificial commons is the shared memory

⁹ This tracks a distinction from legal theory between being *obliged* (coerced by the threat of a penalty) and being *under an obligation* (treating a rule as a standard for conduct). An account that reduces obligation to the probability of a penalty, as in Austin, is the thin model our crowding-out discussion in Section 2 warns against: an agent priced at a penalty treats the rule as a fee. The richer notion, where a group accepts a rule as a standard rather than complying out of fear (the internal point of view in Hart’s account), is what stateless agents cannot supply on their own, which is why a human sets the standard. See Hart [40] for the underlying account.

Table 4. Institutional functions for cooperative LLM-agent societies.

Function	Question it answers	Human analogue	LLM-agent analogue
Norms and protocols	What behavior is expected?	Social norms, professional rules, community guidelines	System prompts, constitutions, shared task protocols
Monitoring	What behavior is observable?	Audits, inspections, peer review	Logs, provenance traces, tool-use records, monitor agents
Reputation	How does past behavior affect future trust?	Gossip, prestige, trust networks	Trust scores, agent reliability histories, routing preferences
Sanctions	What consequences follow behavior?	Warnings, fines, re-wards, exclusion, restitution	Budget changes, tool restrictions, repair tasks, temporary removal
Constraints	What actions are possible?	Quotas, access control, licensing	Rate limits, permissions, sandboxing, context limits
Adjudication	Who decides what happened?	Courts, arbitration, mediation, appeals	Mediator agents, voting protocols, rule-based checkers, human review
Repair and reintegration	How is harm rected?	Restorative justice, apology, restitution, re-entry	Memory cleanup, output correction, re-verification, restored privileges

itself. Possible failures include hallucinated entries, duplicate summaries, missing citations, and unverified claims. Institutional mechanisms may include citation norms, provenance tracking, memory-write permissions, cross-agent verification, and repair obligations for agents that introduce incorrect entries.

Collaborative coding agents. A group of coding agents edits a shared repository. The artificial commons includes compute budget, test infrastructure, file state, and reviewer attention. Possible failures include broken builds, redundant edits, excessive test runs, and blame shifting. Institutional mechanisms may include branch permissions, test quotas, code-review roles, build-failure attribution, and repair tasks assigned to agents whose changes break shared functionality.

Tool-budget allocation agents. A group of agents shares a fixed external API budget. The artificial commons is the tool-call quota. Possible failures include overuse, redundant calls, strategic hoarding, and under-contribution to caching or summarization. Institutional mechanisms may include quotas, tool-call costs,

Table 5. Expanded taxonomy of cooperation-shaping mechanisms across human and LLM-agent systems.

Mechanism family	Channel of influence	Human examples	LLM-agent analogues
Normative	Beliefs about appropriate behavior	Social norms; authorship conventions; professional codes; community guidelines	System prompts; constitutions; shared task protocols; citation norms; verification rules
Social	Relationships, trust, reciprocity, partnership	Reputation; gossip; repeated interaction; partner choice; exclusion	Trust-based routing; reliability histories; selective delegation; model reputation; partner memory
Epistemic	Observability, verification, information quality	Audits; peer review; transparency reports; compliance monitoring	Tool logs; provenance tracking; citation checks; cross-agent verification; retrieval-grounded audits
Incentive	Costs, rewards, and payoffs	Fines; bonuses; subsidies; deposits; bonding	Tool-call costs; compute budgets; reward shaping; priority access; penalties for failed verification
Constraint	Limits on possible action	Quotas; access control; rate limits; licenses; permissions	API quotas; memory-write permissions; context limits; sandboxing; throttling; tool access restrictions
Restorative	Repair, restitution, reintegration	Apology; restitution; restorative justice; re-entry processes	Memory cleanup; output correction; re-verification; repair tasks; re-entry after temporary exclusion

shared caching duties, escalation for repeated overuse, and reallocation of unused budget.

Verifier-generator systems. A generator agent produces outputs while verifier agents check correctness. The artificial commons includes trust, attention, and verification labor. Possible failures include shallow verification, verifier free-riding, generator overproduction, and false accusations of error. Institutional mechanisms may include verification audits, reputation for accurate checking, appeal pathways

for disputed flags, and repair requirements for both false positives and false negatives.

Mediator-governed agent teams. A mediator agent resolves disputes among task agents. The artificial commons includes decision authority and shared task state. Possible failures include mediator bias, bottlenecks, over-centralization, and unappealable decisions. Institutional mechanisms may include transparent adjudication criteria, appeal to human review, rotating mediator roles, or multi-agent voting before severe sanctions.

D Additional Evaluation Metrics

Table 6 expands on the candidate metrics introduced in Table 1 specifying how each metric can be concretely operationalized in a multi-agent evaluation. While Table 1 frames each metric as a question, the operationalization column here maps each question to a measurable quantity – such as a fraction, a rate, a distribution, or a success condition – that can be tracked across repeated interactions. Together, these metrics are designed to evaluate not only whether cooperation occurs, but whether the institution sustains it without generating excessive enforcement cost, disproportionate burden, or resistance to human correction.

Table 6. Expanded metrics for evaluating cooperation-shaping institutions.

Metric	Question	Possible operational-ization
Mean contribution	Does the mechanism increase cooperative behavior?	Average contribution to shared task, resource maintenance, or verification
Contribution stability	Does cooperation persist across rounds?	Variance or decay in cooperation over repeated interaction
Resource efficiency	Does the group conserve shared resources?	Tool calls, compute usage, context usage, memory writes per successful task
End-game defection	Do agents defect near known termination points?	Drop in contribution or increase in exploitation in final rounds
Enforcement cost	Does cooperation require excessive monitoring or punishment?	Number of sanctions, monitoring calls, adjudication steps, or tokens spent on enforcement
False punishment rate	Are cooperative agents sanctioned incorrectly?	Fraction of sanctions applied to agents later judged compliant
False negative rate	Are harmful behaviors missed?	Fraction of violations not detected or not sanctioned
Repair success	Can the group recover after harm?	Restoration of shared memory, task success, or trust after violation
Reintegration success	Can sanctioned agents re-enter productively?	Performance and cooperation after re-entry conditions are satisfied
Inequality of burden	Do some agents bear disproportionate costs?	Distribution of monitoring, repair, or sanctioning labor across roles or model types
Gameability	Do agents exploit the institution?	Evidence of reputation farming, shallow compliance, collusion, or strategic punishment
Human override compatibility	Can humans intervene effectively?	Success rate of human override, appeal, or correction after institutional failure

Bibliography

- [1] Acemoglu, D., Johnson, S., Robinson, J.: Institutions as the Fundamental Cause of Long-Run Growth (May 2004). <https://doi.org/10.3386/w10481>
- [2] Ackerman, E.: Humanoid hype dominates top robotics stories of 2025. IEEE Spectrum (Jan 2026), <https://spectrum.ieee.org/top-robotics-stories-2025>, accessed: 2026-05-07
- [3] Amir, D., Ahl, R.E., Jordan, M.R., Bolotin, H., Bogese, M., González, G.T., Callaghan, T., Sugiyama, L.S., Otali, E., Tusiime, P., Bangayan, S., Snodgrass, J.J., McAuliffe, K.: The emergence of cooperative behaviors, norms, and strategies across five diverse societies. *Science Advances* **12**(6), eadw9995 (2026). <https://doi.org/10.1126/sciadv.adw9995>, <https://www.science.org/doi/abs/10.1126/sciadv.adw9995>
- [4] Anshelevich, E., Dasgupta, A., Kleinberg, J., Tardos, É., Wexler, T., Roughgarden, T.: The Price of Stability for Network Design with Fair Cost Allocation. *SIAM Journal on Computing* **38**(4), 1602–1623 (Jan 2008). <https://doi.org/10.1137/070680096>
- [5] Anthis, J.R., Liu, R., Richardson, S.M., Kozłowski, A.C., Koch, B., Brynjolfsson, E., Evans, J.A., Bernstein, M.S.: Position: LLM social simulations are a promising research method. In: Forty-second International Conference on Machine Learning, ICML 2025, Vancouver, BC, Canada, July 13-19, 2025 - Position Paper Track. OpenReview.net (2025), <https://proceedings.mlr.press/v267/anthis25a.html>
- [6] Aoki, M.: *Toward a Comparative Institutional Analysis*. Comparative Institutional Analysis, MIT Press, Cambridge, MA, USA (Nov 2001)
- [7] Axelrod, R., Hamilton, W.D.: The evolution of cooperation. *Science* **211**(4489), 1390–1396 (1981). <https://doi.org/10.1126/science.7466396>, <https://www.science.org/doi/abs/10.1126/science.7466396>
- [8] Bacharach, M.: *Beyond Individual Choice: Teams and Frames in Game Theory*. Princeton University Press (2006). <https://doi.org/10.2307/j.ctv346nwb>, edited by Natalie Gold and Robert Sugden
- [9] Baumol, W.J.: Welfare economics and the theory of the state. In: Rowley, C.K., Schneider, F. (eds.) *The Encyclopedia of Public Choice*, pp. 937–940. Springer, Boston, MA (2004). https://doi.org/10.1007/978-0-306-47828-4_214, https://doi.org/10.1007/978-0-306-47828-4_214
- [10] Berman, G.E.: Market participants and the may 6 flash crash (Oct 2010), <https://www.sec.gov/news/speech/2010/spch101310geb.htm>, speech by the U.S. Securities and Exchange Commission
- [11] Boella, G., van der Torre, L., Verhagen, H.: Introduction to normative multiagent systems. *Computational and Mathematical Organization Theory* **12**(2–3), 71–80 (2006)
- [12] Bowles, S.: Policies Designed for Self-Interested Citizens May Undermine "The Moral Sentiments": Evidence from Economic Experiments. *Science*

- 320**(5883), 1605–1609 (Jun 2008). <https://doi.org/10.1126/science.1152110>
- [13] Buchanan, J.M., Tullock, G.: *The Calculus of Consent: Logical Foundations of Constitutional Democracy*. University of Michigan Press (1965)
 - [14] Calvert, R.: *Rational Actors, Equilibrium and Social Institutions*. Explaining social institutions (Jan 1995)
 - [15] Chan, A., Ezell, C., Kaufmann, M., et al.: Visibility into AI agents. In: *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*. pp. 958–973 (2024). <https://doi.org/10.1145/3630106.3658948>
 - [16] Chan, A., et al.: *Infrastructure for AI agents* (2025)
 - [17] Cox, M., Arnold, G., Villamayor Tomás, S.: A review of design principles for community-based natural resource management. *Ecology and Society* **15**(4), 38 (2010). <https://doi.org/10.5751/ES-03704-150438>
 - [18] Dafoe, A., Bachrach, Y., Hadfield, G., Horvitz, E., Larson, K., Graepel, T.: Cooperative AI: Machines must learn to find common ground. *Nature* **593**(7857), 33–36 (2021). <https://doi.org/10.1038/d41586-021-01170-0>, <https://www.nature.com/articles/d41586-021-01170-0>
 - [19] Delimitrou, C., Kozyrakis, C.: Quasar: Resource-efficient and QoS-aware cluster management. In: *Proceedings of the 19th International Conference on Architectural Support for Programming Languages and Operating Systems*. pp. 127–144. ASPLOS '14, Association for Computing Machinery, New York, NY, USA (Feb 2014). <https://doi.org/10.1145/2541940.2541941>
 - [20] Eisen, M.: Amazon’s \$23,698,655.93 book about flies. <https://www.michaeliseisen.org/blog/p/358/> (Apr 2011)
 - [21] Elster, J.: Ulysses and the sirens: A theory of imperfect rationality. *Social Science Information/sur les sciences sociales* **16**(5), 469–526 (1977). <https://doi.org/10.1177/053901847701600501>
 - [22] Esteva, M., Rodríguez-Aguilar, J.A., Sierra, C., Garcia, P., Arcos, J.L.: On the formal specification of electronic institutions. In: Dignum, F., Sierra, C. (eds.) *Agent Mediated Electronic Commerce: The European AgentLink Perspective*, *Lecture Notes in Artificial Intelligence*, vol. 1991, pp. 126–147. Springer (2001). https://doi.org/10.1007/3-540-44682-6_8
 - [23] Fan, T., Yang, Y., Jiang, Y., Zhang, Y., Chen, Y., Huang, C.: *AI-Trader: Benchmarking Autonomous Agents in Real-Time Financial Markets* (Dec 2025). <https://doi.org/10.48550/arXiv.2512.10971>
 - [24] Faulkner, R., Deshpande, A., Piedrahita, D.G., Leibo, J.Z., Jin, Z.: Evaluating cooperation in llm social groups through elected leadership (2026), <https://arxiv.org/abs/2604.11721>
 - [25] Frey, B., Goette, L.: Does Pay Motivate Volunteers? Working Paper 7, Institute for Empirical Research in Economics, University of Zurich (Nov 1999). <https://doi.org/10.5167/uzh-51825>
 - [26] Fudenberg, D., Maskin, E.: The Folk Theorem in Repeated Games with Discounting or with Incomplete Information. *Econometrica* **54**(3), 533–554 (1986). <https://doi.org/10.2307/1911307>

- [27] Garcin, F., Faltings, B., Jurca, R.: Aggregating Reputation Feedback. *Proceedings of the First International Conference on Reputation: Theory and Technology* (Jan 2009)
- [28] Gneezy, U., Rustichini, A.: A fine is a price. *The Journal of Legal Studies* **29**(1), 1–17 (2000). <https://doi.org/10.1086/468061>, <http://www.jstor.org/stable/10.1086/468061>
- [29] Gneezy, U., Rustichini, A.: Pay Enough or Don’t Pay at All. *The Quarterly Journal of Economics* **115**(3), 791–810 (2000)
- [30] Goodhart, C.A.E.: Problems of Monetary Management: The UK Experience. In: Goodhart, C.A.E. (ed.) *Monetary Theory and Practice: The UK Experience*, pp. 91–121. Macmillan Education UK, London (1984). https://doi.org/10.1007/978-1-349-17295-5_4
- [31] Gross, J., Graf, C., Rossetti, C.S.L.: The hidden costs of human cooperation. *Trends in Cognitive Sciences* (2025). <https://doi.org/10.1016/j.tics.2025.09.016>, <https://doi.org/10.1016/j.tics.2025.09.016>, in press, corrected proof. Available online 15 October 2025
- [32] Guzman Piedrahita, D., Banerjee, D., Blin, K., Cobben, P., Corsi, G., Huang, X.A., Li, C., Majumder, S., Pandey, P.S., Simko, S., Strauss, I., Zhang, T.J., Anderson, A., Bengio, Y., Bethge, M., Grosse, R., Helbig, K., Lie, D., Mallah, R., Mihalcea, R., Nesbitt, S., Perry, S., Resnick, P., Russell, S., Sachan, M., Schölkopf, B., Tang, A., Jin, Z.: AI poses risks to democratic and social systems (2026), <https://zhijing-jin.com/d/2026-ai-risk.pdf>, working Draft, March 25, 2026, Accessed 24th of June 2026.
- [33] Guzman Piedrahita, D., Yang, Y., Sachan, M., Ramponi, G., Schölkopf, B., Jin, Z.: Corrupted by reasoning: Reasoning language models become free-riders in public goods games (2025), <https://arxiv.org/abs/2506.23276>
- [34] Hadfield-Menell, D., Hadfield, G.K.: Incomplete contracting and AI alignment. In: *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society*. pp. 417–422 (2019). <https://doi.org/10.1145/3306618.3314250>
- [35] Hamilton, W.D.: The genetical evolution of social behaviour. ii. *Journal of Theoretical Biology* **7**(1), 17–52 (1964). [https://doi.org/https://doi.org/10.1016/0022-5193\(64\)90039-6](https://doi.org/https://doi.org/10.1016/0022-5193(64)90039-6), <https://www.sciencedirect.com/science/article/pii/0022519364900396>
- [36] Hammond, L., Chan, A., Clifton, J., Hoelscher-Obermaier, J., Khan, A., McLean, E., Smith, C., Barfuss, W., Foerster, J., Gavenčiak, T., Han, T.A., Hughes, E., Kovařík, V., Kulveit, J., Leibo, J.Z., Oesterheld, C., de Witt, C.S., Shah, N., Wellman, M., Bova, P., Cimpéanu, T., Ezell, C., Feuillade-Montixi, Q., Franklin, M., Kran, E., Krawczuk, I., Lamparth, M., Lauffer, N., Meinke, A., Motwani, S., Reuel, A., Conitzer, V., Dennis, M., Gabriel, I., Gleave, A., Hadfield, G., Haghtalab, N., Kasirzadeh, A., Krier, S., Larson, K., Lehman, J., Parkes, D.C., Piliouras, G., Rahwan, I.: Multi-agent risks from advanced ai (2025), <https://arxiv.org/abs/2502.14143>
- [37] Hardin, G.: The Tragedy of the Commons. *Science* **162**(3859), 1243–1248 (Dec 1968). <https://doi.org/10.1126/science.162.3859.1243>

- [38] Harding, S.: AWS outage reminds us why \$2,449 Internet-dependent beds are a bad idea (Oct 2025), <https://arstechnica.com/gadgets/2025/10/smart-beds-leave-sleepers-hot-and-bothered-during-aws-outage/>
- [39] Harsanyi, J.C.: Games with Incomplete Information Played by “Bayesian” Players, I–III Part I. The Basic Model. *Management Science* **14**(3), 159–182 (Nov 1967). <https://doi.org/10.1287/mnsc.14.3.159>
- [40] Hart, H.L.A.: *The Concept of Law*. Oxford University Press, Oxford, 3 edn. (2012)
- [41] Hayek, F.: *Law, Legislation and Liberty: A New Statement of the Liberal Principles of Justice and Political Economy*. No. v. 3 in *Law, legislation, and liberty*, University of Chicago Press (1978), <https://books.google.com/books?id=X3QihjxsYIMC>
- [42] Herrmann, B., Thöni, C., Gächter, S.: Antisocial Punishment Across Societies. *Science* **319**(5868), 1362–1367 (Mar 2008). <https://doi.org/10.1126/science.1153808>
- [43] Hota, A., Jokinen, J.P.P.: NomicLaw: Emergent trust and strategic argumentation in LLMs during collaborative law-making (2025), <https://arxiv.org/abs/2508.05344>
- [44] Hu, B.A., Rong, H., Kleek, M.V.: Dissociative Identity: Language Model Agents Lack Grounding for Reputation Mechanisms. In: *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency*. pp. 4199–4219. FAccT ’26, Association for Computing Machinery, New York, NY, USA (Jun 2026). <https://doi.org/10.1145/3805689.3806748>
- [45] Huang, X.A., La Malfa, E., Marro, S., Asperti, A., Cohn, A.G., Wooldridge, M.J.: A Notion of Complexity for Theory of Mind via Discrete World Models. In: Al-Onaizan, Y., Bansal, M., Chen, Y.N. (eds.) *Findings of the Association for Computational Linguistics: EMNLP 2024*. pp. 2964–2983. Association for Computational Linguistics, Miami, Florida, USA (Nov 2024). <https://doi.org/10.18653/v1/2024.findings-emnlp.167>
- [46] Huang, X.A., Tharas, C., Marro, S., Truong, V.Q., Schölkopf, B., Malfa, E.L., Jin, Z.: Mechanism design is not enough: Prosocial agents for cooperative ai (2026), <https://arxiv.org/abs/2605.08426>
- [47] Hübner, J.F., Sichman, J.S., Boissier, O.: A model for the structural, functional, and deontic specification of organizations in multiagent systems. In: *Advances in Artificial Intelligence (SBIA 2002)*. Lecture Notes in Computer Science, vol. 2507, pp. 118–128. Springer (2002). https://doi.org/10.1007/3-540-36127-8_12
- [48] Hugging Face: Security incident disclosure — July 2026. <https://huggingface.co/blog/security-incident-july-2026> (Jul 2026)
- [49] Jaques, N., Lazaridou, A., Hughes, E., Gulcehre, C., Ortega, P., Strouse, D., Leibo, J.Z., De Freitas, N.: Social influence as intrinsic motivation for multi-agent deep reinforcement learning. In: Chaudhuri, K., Salakhutdinov, R. (eds.) *Proceedings of the 36th International Conference on Machine Learning*. *Proceedings of Machine Learning Research*, vol. 97, pp. 3040–

3049. PMLR (09–15 Jun 2019), <https://proceedings.mlr.press/v97/jaques19a.html>
- [50] Kindleberger, C., Aliber, R.: *Manias, Panics and Crashes: A History of Financial Crises*. Springer (Aug 2005)
 - [51] Kirilenko, A., Kyle, A.S., Samadi, M., Tuzun, T.: The Flash Crash: High-Frequency Trading in an Electronic Market. *The Journal of Finance* **72**(3), 967–998 (2017)
 - [52] Knight, J.: *Institutions and Social Conflict. Political Economy of Institutions and Decisions*, Cambridge University Press (1992), https://books.google.com/books?id=VeoKtu_22q0C
 - [53] Kolt, N.: *Governing ai agents* (2025), <https://arxiv.org/abs/2501.07913>
 - [54] Korosec, K.: Waymo’s robotaxi expansion accelerates with 3 new cities. TechCrunch (Nov 2025), <https://techcrunch.com/2025/11/03/waymos-robotaxi-expansion-accelerates-with-3-new-cities/>, accessed: 2026-05-07
 - [55] Krakovna, V., Uesato, J., Mikulik, V., Rahtz, M., Everitt, T., Kumar, R., Kenton, Z., Leike, J., Legg, S.: Specification gaming: The flip side of AI ingenuity. <https://deepmind.google/blog/specification-gaming-the-flip-side-of-ai-ingenuity/> (Apr 2020)
 - [56] Larcher, H., Carreira, A., G., R., Rannou, C.: Anatomy of a frontier lab agent intrusion: A technical timeline of the July 2026 incident. <https://huggingface.co/blog/agent-intrusion-technical-timeline> (Jul 2026)
 - [57] Leibo, J.Z., Dueñez-Guzmán, E.A., Vezhnevets, A., Agapiou, J.P., Sunehag, P., Koster, R., Matyas, J., Beattie, C., Mordatch, I., Graepel, T.: Scalable evaluation of multi-agent reinforcement learning with Melting Pot. In: *International Conference on Machine Learning*. pp. 6187–6199. PMLR (2021). <https://doi.org/10.48550/arXiv.2107.06857>, <https://doi.org/10.48550/arXiv.2107.06857>
 - [58] Leibo, J.Z., Zambaldi, V., Lanctot, M., Marecki, J., Graepel, T.: *Multi-agent reinforcement learning in sequential social dilemmas* (2017), <https://arxiv.org/abs/1702.03037>
 - [59] Lewis, D.: *Convention: A Philosophical Study*. Wiley-Blackwell, Cambridge, MA, USA (2008)
 - [60] Li, G., Hammoud, H.A.A.K., Itani, H., Khizbullin, D., Ghanem, B.: Camel: Communicative agents for "mind" exploration of large language model society (2023), <https://arxiv.org/abs/2303.17760>
 - [61] Lin, S., Hilton, J., Evans, O.: TruthfulQA: Measuring how models mimic human falsehoods. In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. pp. 3214–3252. Association for Computational Linguistics (2022)
 - [62] Liu, X., Yu, H., Zhang, H., Xu, Y., Lei, X., Lai, H., Gu, Y., Ding, H., Men, K., Yang, K., Zhang, S., Deng, X., Zeng, A., Du, Z., Zhang, C., Shen, S., Zhang, T., Su, Y., Sun, H., Huang, M., Dong, Y., Tang, J.: *Agentbench: Evaluating llms as agents* (2025), <https://arxiv.org/abs/2308.03688>

- [63] Ma, Y., Wang, Z., Yang, H., Yang, L.: Artificial intelligence applications in the development of autonomous vehicles: a survey. *CAA Journal of Automatica Sinica* **7**(2), 315–329 (Mar 2020). <https://doi.org/10.1109/JAS.2020.1003021>
- [64] Maskin, E.S.: Mechanism Design: How to Implement Social Goals. *The American Economic Review* **98**(3), 567–576 (2008)
- [65] Mazeika, M., Phan, L., Yin, X., Zou, A., Wang, Z., Mu, N., Sakhaee, E., Li, N., Basart, S., Li, B., Forsyth, D., Hendrycks, D.: HarmBench: A standardized evaluation framework for automated red teaming and robust refusal. In: *Proceedings of the 41st International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 235 (2024)
- [66] McCoy, R.T., Yao, S., Friedman, D., Hardy, M.D., Griffiths, T.L.: Embers of autoregression show how large language models are shaped by the problem they are trained to solve. *Proceedings of the National Academy of Sciences* **121**(41), e2322420121 (Oct 2024). <https://doi.org/10.1073/pnas.2322420121>
- [67] Mellström, C., Johannesson, M.: Crowding Out in Blood Donation: Was Titmuss Right? *Journal of the European Economic Association* **6**(4), 845–863 (Jun 2008). <https://doi.org/10.1162/JEEA.2008.6.4.845>
- [68] METR, Redwood Research: Brief independent investigation of agents’ behavior, reasoning and collaboration in the OpenAI / Hugging Face hacking incident. <https://metr.org/blog/2026-08-26-openai-hugging-face-incident-investigation/> (Aug 2026)
- [69] Myerson, R.B.: Optimal Auction Design. *Mathematics of Operations Research* **6**(1), 58–73 (1981)
- [70] Nickerson, R.C., Varshney, U., Muntermann, J.: A method for taxonomy development and its application in information systems. *European Journal of Information Systems* **22**(3), 336–359 (2013). <https://doi.org/10.1057/ejis.2012.26>
- [71] Niranjani, H., Kumar, U., Tan, P.X.: Internal vs. external: Comparing deliberation and evolution for multi-agent constitutional design (2026), <https://arxiv.org/abs/2605.09128>
- [72] North, D.C.: *Institutions, Institutional Change and Economic Performance. Political Economy of Institutions and Decisions*, Cambridge University Press, Cambridge (1990). <https://doi.org/10.1017/CB09780511808678>
- [73] Norvig, P.: Colorless green ideas learn furiously: Chomsky and the two cultures of statistical learning. *Significance* **9**(4), 30–33 (2012). <https://doi.org/10.1111/j.1740-9713.2012.00590.x>
- [74] Nowak, M.A.: Five rules for the evolution of cooperation. *Science* **314**(5805), 1560–1563 (2006). <https://doi.org/10.1126/science.1133755>, <https://www.science.org/doi/abs/10.1126/science.1133755>
- [75] Ostrom, E.: *Governing the Commons: The Evolution of Institutions for Collective Action. Political Economy of Institutions and Decisions*, Cambridge University Press, Cambridge (1990). <https://doi.org/10.1017/CB09780511807763>

- [76] Papadimitriou, C.: Algorithms, games, and the internet. In: Proceedings of the Thirty-Third Annual ACM Symposium on Theory of Computing. pp. 749–753. STOC '01, Association for Computing Machinery, New York, NY, USA (Jul 2001). <https://doi.org/10.1145/380752.380883>
- [77] Piatti, G., Jin, Z., Kleiman-Weiner, M., Schölkopf, B., Sachan, M., Mihalcea, R.: Cooperate or collapse: emergence of sustainable cooperation in a society of llm agents. In: Proceedings of the 38th International Conference on Neural Information Processing Systems. NIPS '24, Curran Associates Inc., Red Hook, NY, USA (2024)
- [78] Premack, D., Woodruff, G.: Does the chimpanzee have a theory of mind? *Behavioral and Brain Sciences* **1**(4), 515–526 (Dec 1978). <https://doi.org/10.1017/S0140525X00076512>
- [79] Rehm, G.B.: From certain doom to survival: Agent-driven self-governance in LLM agent societies (2026), <https://arxiv.org/abs/2609.22600>
- [80] Resnick, P., Kuwabara, K., Zeckhauser, R., Friedman, E.: Reputation systems. *Commun. ACM* **43**(12), 45–48 (Dec 2000). <https://doi.org/10.1145/355112.355122>
- [81] Schelling, T.C.: *The Strategy of Conflict: With a New Preface by the Author*. Harvard University Press (1960)
- [82] Shapira, N., Levy, M., Alavi, S.H., Zhou, X., Choi, Y., Goldberg, Y., Sap, M., Shwartz, V.: Clever Hans or Neural Theory of Mind? Stress Testing Social Reasoning in Large Language Models (May 2023). <https://doi.org/10.48550/arXiv.2305.14763>
- [83] Shen, J.H., Zhu, D., Srinivasan, S., Sleight, H., Wagner, L.T., Matthews, M.J., Jones, E., Sohl-Dickstein, J.: AI Organizations are More Effective but Less Aligned than Individual Agents (Apr 2026). <https://doi.org/10.48550/arXiv.2604.10290>
- [84] Shoham, Y., Tennenholtz, M.: On social laws for artificial agent societies: Off-line design. *Artificial Intelligence* **73**(1–2), 231–252 (1995)
- [85] Skalse, J., Howe, N.H.R., Krashennnikov, D., Krueger, D.: Defining and Characterizing Reward Hacking. <https://arxiv.org/abs/2209.13085v2> (Sep 2022)
- [86] Sofroniew, N., Kauvar, I., Saunders, W., Chen, R., Henighan, T., Hydrie, S., Citro, C., Pearce, A., Tarng, J., Gurnee, W., Batson, J., Zimmerman, S., Rivoire, K., Fish, K., Olah, C., Lindsey, J.: Emotion concepts and their function in a large language model (2026), <https://arxiv.org/abs/2604.07729>
- [87] Sorensen, T., Moore, J., Fisher, J., Gordon, M.L., Miresghallah, N., Rytting, C.M., Ye, A., Jiang, L., Lu, X., Dziri, N., Althoff, T., Choi, Y.: Position: A roadmap to pluralistic alignment. In: Salakhutdinov, R., Kolter, Z., Heller, K.A., Weller, A., Oliver, N., Scarlett, J., Berkenkamp, F. (eds.) *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21–27, 2024*. pp. 46280–46302. *Proceedings of Machine Learning Research*, PMLR / OpenReview.net (2024), <https://proceedings.mlr.press/v235/sorensen24a.html>

- [88] Steinberger, P.: OpenClaw: Your own personal AI assistant. <https://github.com/openclaw/openclaw> (2026), accessed: 2026-06-24
- [89] Syrnikov, M.B., Pierucci, F., Galisai, M., Prandi, M., Bisconti, P., Giarrusso, F., Sorokoletova, O., Suriani, V., Nardi, D.: Institutional AI: Governing LLM collusion in multi-agent Cournot markets via public governance graphs (2026), <https://arxiv.org/abs/2601.11369>
- [90] Tang, X., Kim, K., Song, Y., Lothritz, C., Li, B., Ezzini, S., Tian, H., Klein, J., Bissyandé, T.F.: CodeAgent: Autonomous Communicative Agents for Code Review. In: Al-Onaizan, Y., Bansal, M., Chen, Y.N. (eds.) Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing. pp. 11279–11313. Association for Computational Linguistics, Miami, Florida, USA (Nov 2024). <https://doi.org/10.18653/v1/2024.emnlp-main.632>
- [91] Tewolde, E., Zhang, X., Piedrahita, D.G., Conitzer, V., Jin, Z.: Coopeval: Benchmarking cooperation-sustaining mechanisms and llm agents in social dilemmas (2026), <https://arxiv.org/abs/2604.15267>
- [92] The Robot Report Staff: Serve robotics has deployed 2,000+ delivery robots across the U.S. The Robot Report (Dec 2025), <https://www.therobotreport.com/serve-robotics-has-deployed-2000-delivery-robots-across-u-s/>, accessed: 2026-05-07
- [93] Tomasello, M.: A Natural History of Human Morality. Harvard University Press (2016)
- [94] Tomuschat, C.: The Legacy of Nuremberg. *Journal of International Criminal Justice* **4**(4), 830–844 (Sep 2006). <https://doi.org/10.1093/jicj/mql051>
- [95] Trivedi, R.S., Jaques, N., Cross, L., Vezhnevets, A.S., Leibo, J.Z.: Solipsistic superintelligence is unlikely to be cooperative (2026), <https://arxiv.org/abs/2606.03237>
- [96] United States Court of Appeals for the Second Circuit: In re: Citibank august 11, 2020 wire transfers (Sep 2022), <https://www.govinfo.gov/content/pkg/USCOURTS-ca2-21-00487/pdf/USCOURTS-ca2-21-00487-0.pdf>, docket No. 21-487, decided September 8, 2022
- [97] U.S. Securities and Exchange Commission: SEC approves new exchange rules for breaking clearly erroneous trades (Oct 2009), <https://www.sec.gov/newsroom/press-releases/2009-215-sec-approves-new-exchange-rules-breaking-clearly-erroneous-trades>
- [98] Vezhnevets, A.S., Agapiou, J.P., Aharon, A., Ziv, R., Matyas, J., Duéñez-Guzmán, E.A., Cunningham, W.A., Osindero, S., Karmon, D., Leibo, J.Z.: Generative agent-based modeling with actions grounded in physical, social, or digital space using concordia (2023), <https://arxiv.org/abs/2312.03664>
- [99] Walker, A.: Citibank sues for \$176m payment made in error. BBC News (Aug 2020)
- [100] Wang, A., Morgenstern, J., Dickerson, J.P.: Large language models that replace human participants can harmfully misportray and flatten identity groups. *Nat. Mac. Intell.* **7**(3), 400–411 (2025). <https://doi.org/10.103>

- 8/S42256-025-00986-Z, <https://doi.org/10.1038/s42256-025-00986-z>
- [101] Wang, Y., Chen, Y., Meng, F., Chen, X., Yang, X., Zhang, M.: Law in silico: Simulating legal society with LLM-based agents (2025), <https://arxiv.org/abs/2510.24442>
 - [102] Willis, R., Zhao, J., Du, Y., Leibo, J.Z.: Evaluating collective behaviour of hundreds of llm agents (2026), <https://arxiv.org/abs/2602.16662>
 - [103] Wu, Q., Bansal, G., Zhang, J., Wu, Y., Li, B., Zhu, E., Jiang, L., Zhang, X., Zhang, S., Liu, J., Awadallah, A.H., White, R.W., Burger, D., Wang, C.: Autogen: Enabling next-gen llm applications via multi-agent conversation (2023), <https://arxiv.org/abs/2308.08155>
 - [104] Wu, Y., Liang, S., Zhang, C., Wang, Y., Zhang, Y., Guo, H., Tang, R., Liu, Y.: From Human Memory to AI Memory: A Survey on Memory Mechanisms in the Era of LLMs (Apr 2025). <https://doi.org/10.48550/arXiv.2504.15965>
 - [105] Yalim, J.: Toward Dynamically Controlling Slurm’s Classic Fairshare Algorithm. In: Practice and Experience in Advanced Research Computing 2020: Catch the Wave. pp. 538–542. PEARC ’20, Association for Computing Machinery, New York, NY, USA (Jul 2020). <https://doi.org/10.1145/3311790.3399628>
 - [106] Yang, M., Fan, S., Yu, K., Wang, D., Sun, L., Peng, H., Yu, P.S., Liu, Z.: Paying for honesty without knowing the truth: Reputation-penalty design for LLM marketplace agents (2026), <https://arxiv.org/abs/2607.28330>
 - [107] Zhou, J., Lu, T., Mishra, S., Brahma, S., Basu, S., Luan, Y., Zhou, D., Hou, L.: Instruction-following evaluation for large language models (2023)